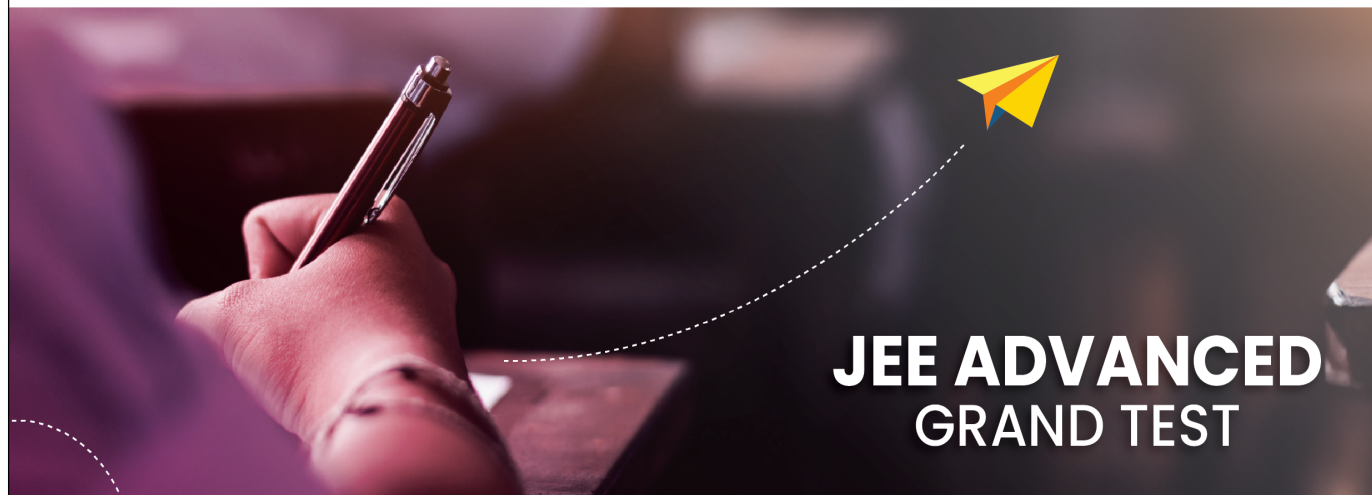




Sri Chaitanya
Educational Institutions



JEE ADVANCED GRAND TEST



Sri Chaitanya IIT Academy.,India.

✦ A.P ✦ T.S ✦ KARNATAKA ✦ TAMILNADU ✦ MAHARASTRA ✦ DELHI ✦ RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: **Sr.Super60_Elite, Target & LIIT-BTs**

Paper -1(Adv-2023-P1-Model)

Date: 08-05-2024

Time: 09.00Am to 12.00Pm

GTA-31

Max. Marks: 180

08-05-2024_**Sr.Super60_Elite,Target & LIIT-BTs**_Jee-Adv(2023-P1)_GTA-31_Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Name of the Student: _____

H.T. NO:

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**JEE-ADVANCE-2023-P1-Model**

Time:3Hr's

IMPORTANT INSTRUCTIONS

Max Marks: 180

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 3)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 4 – 7)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 8 – 13)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				17	60

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 20)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 21 – 24)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 25 – 30)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
Total				17	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 37)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 38 – 41)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 42 – 47)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				17	60

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Space for rough work

Page 2

**Sri Chaitanya**
Educational Institutions**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN
2023**
SINGARAJ
VENKAT KOUNDINNYA
Sri Chaitanya
100th Class
300
300

**RANK
1**

**JEE Advanced
2023**
VARUN
CHIDVILAS REDDY
Sri Chaitanya
100th Class
341
360

**RANK
1**

**NEET
2023**
BODA VARUN
CHAKRAVARTHI
Sri Chaitanya
100th Class
720
720

**RANK
1**

**MATHEMATICS****Max Marks: 60****SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)**This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;**Partial Marks** : +3 If all the four options are correct but **ONLY** three options are chosen;**Partial Marks** : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;**Partial Marks** : +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered);**Negative Marks** : -2 In all other cases.

For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then choosing **ONLY** (A), (B) and (D) will get +4 marks; choosing **ONLY** (A) and (B) will get +2 marks; choosing **ONLY** (A) and (D) will get +2 marks; choosing **ONLY** (B) and (D) will get +2 marks; choosing **ONLY** (A) will get +1 mark; choosing **ONLY** (B) will get +1 mark; choosing **ONLY** (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

- Let $f(x) = (x + |x|)|x|$, then for all x
 - f is continuous
 - f' is differentiable $\forall x \in R$
 - f is differentiable
 - f'' is continuous
- Let 'z' be a complex number and 'a' be a positive real parameter such that $z^2 + az + a^2 = 0$, then
 - locus of z (as 'a' varies) is a pair of rays
 - locus of z (as 'a' varies) is a circle
 - $\arg(z) = \pm \frac{2\pi}{3}$
 - $|z| = |a|$
- Let $AB : y = x$ be a chord to the parabola $y^2 + 4x = 4$ and P be any variable point on the ellipse $x^2 + 9(y - 4)^2 - 9 = 0$. If Δ_1 and Δ_2 be the minimum and maximum area of triangle ABP respectively then
 - $\Delta_2 = 2\sqrt{2}(4 + \sqrt{10})$
 - $\Delta_2 - \Delta_1 = 8\sqrt{5}$
 - $\frac{\Delta_2}{\Delta_1} = \frac{13 + 4\sqrt{10}}{3}$
 - none of these

**SECTION – II
(SINGLE CORRECT ANSWER TYPE)**This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered);**Negative Marks** : -1 In all other casesSec: **Sr.Super60_Elite, Target & LIIT-BTs**

Space for rough work

Page 3

**Sri Chaitanya**
Educational Institutions**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**
**JEE MAIN
2023**
SINHA
VENKAT KOUNDYIN
300
300
RANK 1

**RANK
1**
**JEE Advanced
2023**
VARUN
CHIVILAS REDDY
341
360
RANK 1

**RANK
1**
**NEET
2023**
SODA VARUN
CHAKRAVARTI
720
720
RANK 1

**RANK
1**



4. Let

$g(x) = f_n(x)$, where $f_2(x) = f(f(x))$, $f_3(x) = f(f(f(x)))$and

$f(x) = x^3 - \frac{3}{2}x^2 + x + \frac{1}{4}$ then $\int_{1/4}^{3/4} g(x)dx$ is

- A) $\frac{n}{2} + 2$ B) $2^n + 1$ C) $\frac{1}{4}$ D) $\frac{3}{4}$

5. The vectors $(x^2 - 1)\vec{i} + 2(x^2 - 1)\vec{j} - 3(x^2 - 1)\vec{k}$, $(2x^2 - 1)\vec{i} + (2x^2 + 1)\vec{j} + x^2\vec{k}$, and $(3x^2 + 2)\vec{i} + (x^2 + 4)\vec{j} + (x^2 + 1)\vec{k}$ are non-coplanar. The number of real values that x cannot take is

- A) 0 B) 6 C) 2 D) 4

6. Number of solutions of the equation $[y + [y]] = 2 \cos x$ is, where

$y = \frac{1}{3}[\sin x + [\sin x + [\sin x]]]$ and $[.]$ denotes the greatest integer function

- A) 1 B) 2 C) 3 D) none of these

7. If three one by one squares are selected at random from the chessboard, then the probability that they form the letter L is

- A) $\frac{196}{64C_3}$ B) $\frac{49}{64C_3}$ C) $\frac{36}{64C_3}$ D) $\frac{98}{64C_3}$

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If **ONLY** the correct integer is entered; Zero Marks : 0 In all other cases.

8. Given that $g(x) = 2f\left(\frac{x^2}{2}\right) + f(6 - x^2) \forall x \in R$ and $f^{11}(x) > 0 \forall x \in R$. The set of all points where $g(x)$ is increasing is $(a, b) \cup (c, \infty)$ then $[a + b + c]$ is _____ (where $[.]$ denotes the greatest integer function).

9. A diagonal of rhombus ABCD is a member of both the family of lines $(x + y - 1) + \lambda_1(2x + 3y - 2) = 0$ and $(x - y + 2) + \lambda_2(2x - 3y + 5) = 0$ where $\lambda_1, \lambda_2 \in R$ and one of the vertex of the rhombus is $(3, 2)$. If area of the rhombus is $12\sqrt{5}$ square units then the length of longer diagonal of the rhombus is _____





10. If the quadratic $x^2 + (2 - \tan \theta)x - (1 + \tan \theta) = 0$ has two integral roots, then sum of all possible values of θ in the interval $(0, 2\pi)$ is $k\pi$, then the value of k is equal to _____
11. The number of normal(s) to hyperbola which can touch its conjugate is equal to _____
12. For $y = \cot^{-1} \left(\frac{\log(ex^2)}{\log(e/x^2)} \right) + \tan^{-1} \left(\frac{3 + 2 \ln x}{1 - 6 \ln x} \right)$, the value of $\frac{d^2 y}{dx^2}$ at $x=1$, is equal to _____
13. If a singular matrix $A = [a_{ij}]_{2 \times 2}$ always commutes with $B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ such that $\left| \frac{a_{11}}{a_{12}} \right| = \sqrt{k}$, then k is equal to _____

SECTION – IV (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks: +3 ONLY if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -1 In all other cases.

14. Match the following List-I with List-II

	COLUMN-I		COLUMN-II
P	Let $f(x) = x^2 - xg'(1) + g''(2)$ and $g(x) = x^2 + xf'(2) + f''(3)$, then $2(f'(2) - f'(1))$ is equal to	1.	2
Q	If $f(x-y), f(x), f(y)$ and $f(x+y)$ are in A.P for all x, y and $f(0) \neq 0$, then $f'(2) + f'(-2)$ is equal to	2.	1
R	If $f(x) = x^3 + x^2.f'(1) + xf''(2) + f'''(3)$ for all x , then $f(0) + f(3) + 1$ is equal to	3.	4
S	Let $f(x) = x^n$, n being a positive integer, then value of n for which the equality $f'(a+b) = f'(a) + f'(b)$ is valid for all $a, b > 0$ is	4.	0

A) P-4; Q-3; R-2; S-1

B) P-3; Q-4; R-1; S-2

C) P-3; Q-4; R-2; S-1

D) P-1; Q-2; R-3; S-4

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Space for rough work

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**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN
2023**
SINHAJU
VENKAT KOUNDURU
Sri Chaitanya
All India Rank
300
300



**RANK
1**

**JEE Advanced
2023**
VINOOD
CHIDVILAS REDDY
Sri Chaitanya
All India Rank
341
360



**RANK
1**

**NEET
2023**
BORA VARUN
CHAKRAVARTI
Sri Chaitanya
All India Rank
720
720



**RANK
1**



15. A function F is defined by $F(x) = \int_1^x \frac{e^t}{t} dt \forall x > 0$. Now express the function in List-I in terms of F . Then match the following List-I with List-II

	COLUMN-I		COLUMN-II
P	$\int_1^x \frac{e^t}{t+2} dt$	1.	$F(x) - \frac{e^x}{x} + e$
Q	$\int_1^x \frac{e^{3t}}{t} dt$	2.	$xe^x - e - F\left(\frac{1}{x}\right)$
R	$\int_1^x \frac{e^t}{t^2} dt$	3.	$e^{-2} [F(x+2) - F(3)]$
S	$\int_1^x e^{\frac{1}{t}} dt$	4.	$F(3x) - F(3)$

- A) P-4; Q-3; R-2; S-1 B) P-3; Q-4; R-1; S-2
C) P-3; Q-4; R-2; S-1 D) P-1; Q-2; R-3; S-4

16. Match the following List-I with List-II

	COLUMN-I		COLUMN-II
P	An urn contains five balls, two balls are drawn and are found to be white. If probability of all the balls in urn are white is k , then	1.	$k = \frac{31}{91}$
Q	Out of 15 consecutive integers three are selected at random, then the probability of the sum is divisible by 3 is k , then	2.	$k = \frac{3}{16}$
R	If 3 different cards are placed at random and independently in 4 boxes lying in a straight line (one box can have more than one card). Then the probability of the cards going into 3 adjacent boxes is k , then	3.	$k = \frac{3}{4}$
S	A box contains 4 Red balls and 1 black ball. Balls are drawn one by one without replacement. If the 5 th ball drawn is Red then the probability that the 3 rd ball drawn was Red is k , then $k =$	4.	$k = \frac{1}{2}$





A) P- 4; Q-3; R-2; S-1

B) P-3; Q-2; R-4; S-1

C) P-3; Q-4; R-2; S-1

D) P-4; Q-1; R-2; S-3

17. Match the following List-I with List-II

	COLUMN-I		COLUMN-II
P	$\sin \left[\frac{\pi}{900} \left\{ \sum_{r=1}^{10} (r - \omega)(r - \omega^2) \right\} \right]$ is equal to	1.	0
Q	If the roots of $t^2 + t + 1 = 0$ be α, β , then $\alpha^4 + \beta^4 + \alpha^{-1}\beta^{-1}$ is equal to	2.	4
R	If $\left[\frac{1 + \cos \theta + i \sin \theta}{\sin \theta + i(1 + \cos \theta)} \right]^4 = \cos n\theta + i \sin n\theta$, then n is equal to	3.	i
S	If $z_r = \cos \frac{\pi}{3^r} + i \sin \frac{\pi}{3^r}$ (where $r=1,2,3,\dots$), then value of $z_1 z_2 z_3 \dots \infty$ is equal to	4.	1

A) P-4; Q-1; R-3; S-2

B) P-1; Q-4; R-3; S-2

C) P-2; Q-3; R-4; S-1

D) P-4; Q-1; R-2; S-3



PHYSICS

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;

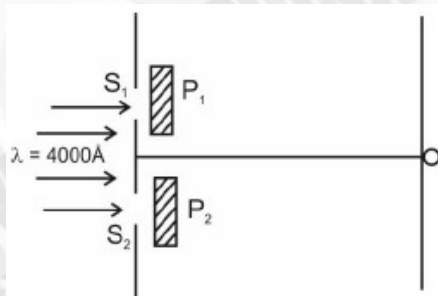
Partial Marks : +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

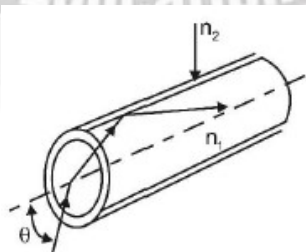
Negative Marks : -2 In all other cases.

For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then choosing **ONLY** (A), (B) and (D) will get +4 marks; choosing **ONLY** (A) and (B) will get +2 marks; choosing **ONLY** (A) and (D) will get +2 marks; choosing **ONLY** (B) and (D) will get +2 marks; choosing **ONLY** (A) will get +1 mark; choosing **ONLY** (B) will get +1 mark; choosing **ONLY** (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

18. P_1 and P_2 are transparent plates having equal thickness $20\mu\text{m}$ and refractive indices $\mu_1 = 1.7$ $\mu_2 = 1.6$. P_1 transmits 75% whereas P_2 transmits 50% of energy incident. [Given S_1 and S_2 are identical slits and the intensity at 'O' without plates is $I = 4I_0$]



- A) The optical path difference between waves coming from S_1 and S_2 at 'O' is $\Delta x = 4 \times 10^{-6} \text{ m}$
- B) The phase difference between waves coming from S_1 and S_2 at 'O' is $\Delta\phi = 10\pi$
- C) The intensity at 'O' after placing P_1 and P_2 is $I = \frac{5I_0}{4} + \sqrt{\frac{3}{2}}I_0$
- D) The intensity at 'O' does not change even after placing plates P_1 and P_2
19. An optical fiber consists of a non-graded glass core (index of refraction n_1) surrounded by a cladding (index of refraction n_2) $n_2 < n_1$. Suppose if a beam of light enters the fiber from air at an angle ' θ ' with the fiber axis as shown. Then (c is speed of light in vacuums)





A) The greatest possible value of ' θ ' for which ray can be propagated through fiber is

$$\theta = \sin^{-1} \sqrt{n_1^2 - n_2^2}$$

B) The greatest possible value of ' θ ' for which ray can be propagated through fiber is

$$\theta = \cos^{-1} \sqrt{n_1^2 - n_2^2}$$

C) The time difference between ray that travels ' L ' distance along fiber axis and that is reflecting at critical angle is $\Delta t = \frac{L n_2}{c n_1} (n_1 - n_2)$

D) The time difference between ray that travels ' L ' distance along fiber axis and that is reflecting at critical angle is $\Delta t = \frac{L n_1}{c n_2} (n_1 - n_2)$

20. A small block of mass m and charge $+q$ is kept on a smooth horizontal xy -plane. Both electric field and magnetic field are acting in the region given by $\vec{E} = E_0 \hat{i} N C^{-1}$ and $\vec{B} = B_0 \hat{j}$ respectively at $t = 0$. Block starts moving on the plane with zero velocity, then [Gravity acts along negative Z -direction]

A) Block remains at rest at any instant

B) Block leaves the contact when speed of the block is $\frac{mq}{qB_0}$

C) After some time block moves with constant velocity

D) Block leaves the contact at an instant $t = \frac{m^2 g}{q^2 E_0 B_0}$

SECTION – II (SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

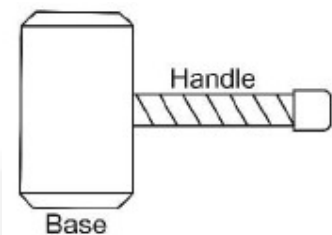
Negative Marks : -1 In all other cases

21. If the wavelength of the incident radiation is increased from 300 nm to 301 nm, then find the corresponding change in the stopping potential

A) $\Delta V = -0.674 V$ B) $\Delta V = -0.487 V$ C) $\Delta V = -0.272 V$ D) $\Delta V = -0.0137 V$

22. If R and r are the radius of base and handle of a hammer which can sustain a maximum shear stress of ' S ' pascals, then the maximum force with which we can hit any object with hammer, is





- A) $\pi S r^2$ B) $\pi S R^2$ C) $\pi S(r^2 + R^2)$ D) $\frac{\pi S r^4}{R^2}$

23. A rope thrown over a pulley has a ladder with a man of mass m on one of its ends and a counter balancing mass M on its other end. The man climbs with a velocity V_r relative to ladder. Ignoring the masses of pulley and the rope as well as the friction on the pulley axis, the velocity of the centre of mass of this system is

- A) $\frac{m}{M} V_r$ B) $\frac{m}{2M} V_r$ C) $\frac{M}{m} V_r$ D) $\frac{3M}{2m} V_r$

24. A circular disc of radius R and thickness $\frac{R}{6}$ has moment of inertia I about an axis passing through its centre and perpendicular to its plane. It is melted and recasted into a solid sphere. The moment of inertia of the sphere about its diameter as axis of rotation is

- A) I B) $\frac{2I}{8}$ C) $\frac{I}{5}$ D) $\frac{I}{10}$

SECTION-III

(NON-NEGATIVE INTEGER.)

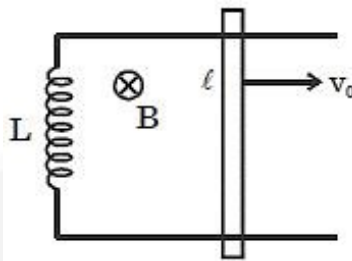
This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

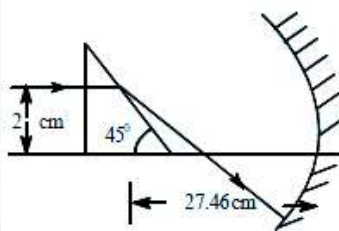
Full Marks : +4 If **ONLY** the correct integer is entered; Zero Marks : 0 In all other cases.

25. A loop is formed by two parallel conductors connected by a solenoid with inductance $L = 2$ Hand a conducting rod of mass $m = 80\text{g}$, which can move freely (without friction) and slide over the conductors. The conductors are located in a horizontal plane with a uniform vertical magneticfield $B = 4\text{T}$ as shown in figure. The distance between two parallel conductors is equal to $\ell = 4\text{cm}$. At the moment $t = 0$, the conducting rod is imparted an initial velocity $v_0 = 2\text{ cm/s}$ directed to the right. The amplitude (in cm) for the oscillations of conducting rod.

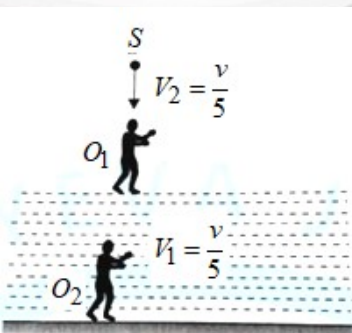




26. A ray of light falls normally (as shown) on a right angled prism whose $\mu = \sqrt{1.5}$ and 45° . After passing through prism, it falls on a concave mirror which is at a distance of 27.46 cm from the point of emergence of prism. The net deviation of the ray is $n \times 90^\circ$, (if focal length of the mirror is 10 cm) then value of n is

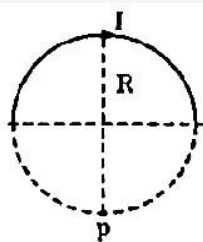


27. Nuclei A and B convert into a stable nucleus C. Nucleus A is converted into C by emitting two α particles and three β particles. Nucleus B is converted into C by one α particles and five β particles. At time $t = 0$, nuclei of A are $4N_0$ and nuclei of B are N_0 . Initially, number of nuclei of C are zero. Half-life of A (into conversion of C) is 1 min and that of B is 2 min. Find the time (in minutes) at which rate of disintegration of A and B are equal.
28. In the figure shown, an observer O_1 floats (static) on water surface with ears in air while another observer O_2 is moving upwards with constant velocity $V_1 = V/5$ in water. The source moves down with constant velocity $V_2 = V/5$ and emits sound of frequency f . The velocity of sound in air is V and that in water is $4V$. For this situation the wavelength of the sound received by O_2 is $\frac{16V}{xf}$. Find x





29. A thin wire bent into the form of a semicircle of radius R carries a current I . The magnetic field of induction at the point P as shown in the figure is $\frac{\mu_0 I}{4\pi R} \ln(\sqrt{n} + 1)$, then the value of n is _____



30. An equi-molar mixture of a mono atomic and diatomic gas is subjected to continuous expansion such that $dQ = \frac{1}{3}dU + \frac{1}{2}dW$. The equation of the process in terms of the variable P & V is $PV^{11/x} = \text{constant}$. Then value of x is _____.

SECTION – IV (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 ONLY if the option corresponding to the correct combination is chosen;

Zero Marks: 0 if none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -1 in all other cases.

31. In each situation of List-I, a process $A \rightarrow B \rightarrow C$ is given for an ideal gas. Match each situation of List –I with correct result in List -II.

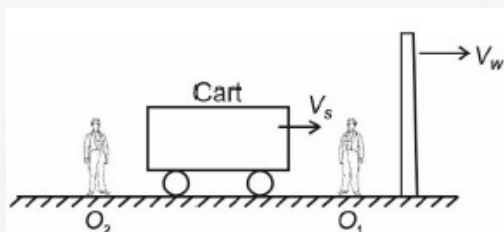
List – I	List - II
(P)	(1) Temperature first decreases and then increases
(Q)	(2) Temperature increases continuously
(R)	(3) Pressure first increases and then remain constant
(S)	(4) Pressure first decreases and then remain constant





- A) $P \rightarrow (2,4), Q \rightarrow (1,3), R \rightarrow (3), S \rightarrow (3)$
 B) $P \rightarrow (2,3), Q \rightarrow (1,4), R \rightarrow (2,3), S \rightarrow (3)$
 C) $P \rightarrow (2,4), Q \rightarrow (1,4), R \rightarrow (3,1), S \rightarrow (3)$
 D) $P \rightarrow (2,3), Q \rightarrow (1,3), R \rightarrow (3), S \rightarrow (3)$

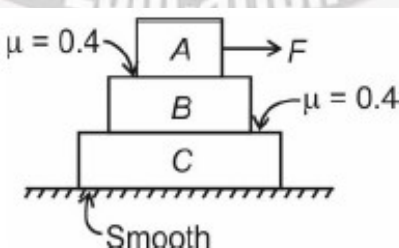
32. Observer O_1 and O_2 are at rest and wall is moving with speed V_w . Speed of car is V_s and emits frequency f_0 . Speed of sound with respect to ground is V . Then match the List-I with List-II



List – I		List - II	
(P)	Wavelength received by O_1 directly from cart	(1)	$\frac{V + V_s}{f_0}$
(Q)	Wavelength received by O_2 directly from cart	(2)	$\frac{V - V_s}{f_0}$
(R)	Wavelength received by driver of the cart after reflection from wall	(3)	$\left(\frac{V + V_w}{V - V_w} \right) \left(\frac{V - V_s}{f_0} \right)$
(S)	Wavelength received by O_2 after reflection from wall	(4)	$\left(\frac{V + V_w}{V - V_w} \right) \left(\frac{V + V_s}{f_0} \right)$

- A) $P \rightarrow (2), Q \rightarrow (1), R \rightarrow (3), S \rightarrow (4)$ B) $P \rightarrow (1), Q \rightarrow (2), R \rightarrow (4), S \rightarrow (3)$
 C) $P \rightarrow (2), Q \rightarrow (1), R \rightarrow (3), S \rightarrow (3)$ D) $P \rightarrow (1), Q \rightarrow (2), R \rightarrow (3), S \rightarrow (4)$

33. Three blocks A, B, C of mass 1 kg, 2 kg and 3 kg are placed on smooth horizontal surface as shown. Coefficient of friction between A, B and C is 0.40. A force F is applied on block A . Force is applied in such a way that system moves together with maximum possible acceleration ($g = 10 \text{ m/s}^2$)

JEE MAIN
2023SIRARAJ
VENKAT KOUNDINNYA
RANK 1
300
300RANK
1JEE Advanced
2023VARUN
CHIDVILAS REDDY
RANK 1
341
360RANK
1NEET
2023BORA VARUN
CHAKRAVARTHI
RANK 1
720
720RANK
1



COLUMN – I		COLUMN - II	
(A)	Force F is	(P)	2.4 N
(B)	Friction force between A and B is	(Q)	4.0 N
(C)	Friction force between B and C is	(R)	4.8 N
(D)	Net force on A is	(S)	0.8 N
		(T)	1.6 N

A) $A \rightarrow (R); B \rightarrow (Q); C \rightarrow (S); D \rightarrow (P)$

B) $A \rightarrow (Q); B \rightarrow (S); C \rightarrow (P); D \rightarrow (R)$

C) $A \rightarrow (Q); B \rightarrow (R); C \rightarrow (P); D \rightarrow (S)$

D) $A \rightarrow (R); B \rightarrow (Q); C \rightarrow (P); D \rightarrow (S)$

34. Column-I shows a situation for placing an object and column-II marks corresponding properties of image. Match the columns

COLUMN – I		COLUMN - II	
(A)	Virtual object kept in concave mirror	(P)	Virtual image is formed
(B)	Virtual object between pole and focus of concave lens, kept in air	(Q)	Real image is formed
(C)	Real object kept in between pole and focus of convex lens immersed in medium of larger refractive index the material of lens	(R)	Enlarged image is formed
(D)	Virtual object kept beyond centre of curvature in convex mirror	(S)	Diminished image is formed
		(T)	Virtual and Inverted image is formed

A) $A \rightarrow (Q, T); B \rightarrow (Q, R); C \rightarrow (P, S); D \rightarrow (P, S)$

B) $A \rightarrow (Q, S); B \rightarrow (Q, R); C \rightarrow (P, S); D \rightarrow (P, S)$

C) $A \rightarrow (P, S); B \rightarrow (Q, S); C \rightarrow (P, T); D \rightarrow (P, R)$

D) $A \rightarrow (Q, S); B \rightarrow (Q, R); C \rightarrow (P, T); D \rightarrow (P, R)$



CHEMISTRY

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

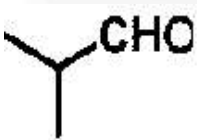
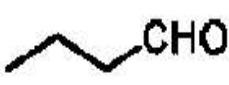
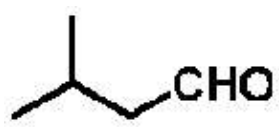
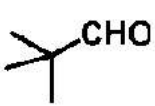
Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;**Partial Marks** : +3 If all the four options are correct but **ONLY** three options are chosen;**Partial Marks** : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;**Partial Marks** : +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered);**Negative Marks** : -2 In all other cases.

For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then choosing **ONLY** (A), (B) and (D) will get +4 marks; choosing **ONLY** (A) and (B) will get +2 marks; choosing **ONLY** (A) and (D) will get +2 marks; choosing **ONLY** (B) and (D) will get +2 marks; choosing **ONLY** (A) will get +1 mark; choosing **ONLY** (B) will get +1 mark; choosing **ONLY** (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

35. For element $X_{(g)}$ value of negative electron gain enthalpy is second highest in periodic table elements. Select **CORRECT** statement(s) for element X.

- A) Electronegativity of X is higher as compared to $Cl_{(g)}$ on Paulings scale
- B) Negative electron gain enthalpy of $X_{(g)}^+$ is more than negative electron gain enthalpy of $O_{(g)}^+$
- C) Negative electron gain enthalpy of $X_{(g)}$ is more than negative electron gain enthalpy of $Cl_{(g)}$
- D) Ionisation energy of $X_{(g)}^-$ is less than ionization of $Cl_{(g)}^-$

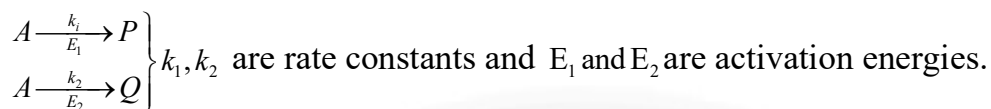
36. An alkyl halide(X) on reaction with ethanolic sodium hydroxide forms an alkene(Y) which on further reaction with HBr gives the same alkyl halide. The alkene(Y) on reaction with HBr/peroxide followed by reaction with Mg metal followed by reaction with HCN produces an aldehyde(Z) (here consider the reaction between $RMgX$ and HCN is nucleophilic addition not the acid base reaction). Among the following Z can be

- A) 
- B) 
- C) 
- D) 





37. Consider the following parallel, first order processes



The correct deduction(s) is (are)

A) At any time, the ratio of concentration of P and Q has a fixed value at a given temperature.

B) If a catalyst is added, it will decrease both E_1 and E_2 by same amount.

C) If 'E' represents the overall activation energy for decay of A, it will be $\left(\frac{E_1 k_1 + E_2 k_2}{k_1 + k_2} \right)$

D) If 'k' represents the overall rate constant for decay of A, it will be $k_1 + k_2$

SECTION – II (SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

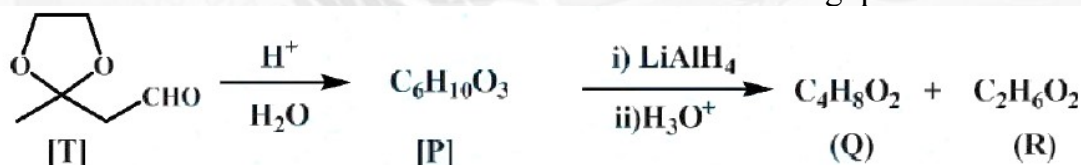
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

38. Observe the scheme of the reaction and answer the following questions



Choose the incorrect statement(s) regarding [P]

A) [P] can respond to the 2,4-DNP test

B) [P] is an isomer of [T]

C) [P] can respond to the fehling test

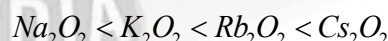
D) [P] can respond to the iodoform test

39. Which is not correct matched?

1) Basic strength of oxides



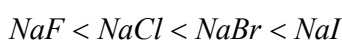
2) Stability of peroxides.



3) Stability of bicarbonates.



4) Melting point



A) 1 and 4

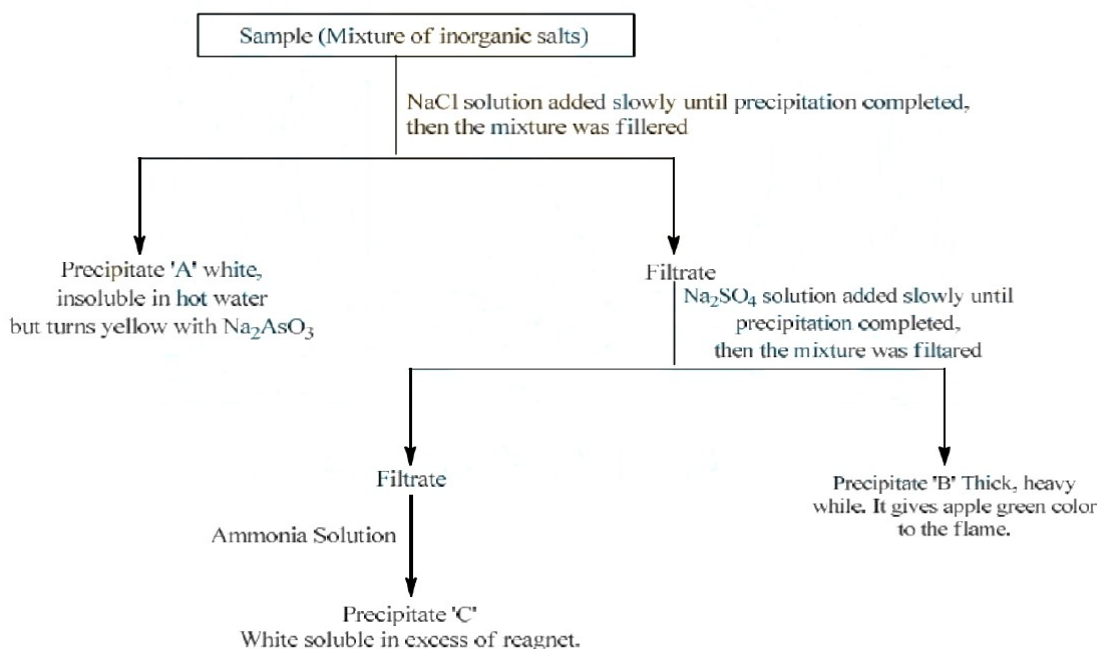
B) 1 and 3

C) 1 and 2

D) 2 and 3

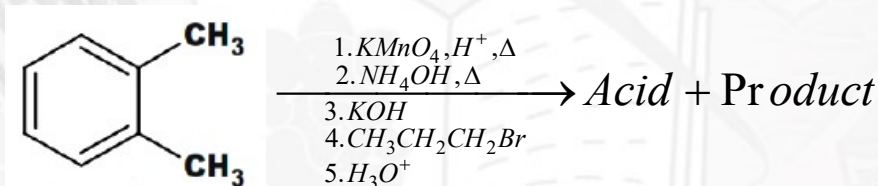


40. A study was given a sample of colourless solution containing three cations and was asked to identify the cations. Student carried out a series of reactions as given below.



Precipitates 'A', 'B' and 'C' are respectively:

- A) $\text{Al}(\text{OH})_3$, BaSO_4 and AgCl B) AgCl , BaSO_4 and $\text{Zn}(\text{OH})_2$
 C) AgCl , $\text{Ca}(\text{OH})_2$ and ZnSO_4 D) ZnCl_2 , BaSO_4 and $\text{Al}(\text{OH})_3$
41. The major product formed in the following reaction is:



- A) $\text{CH}_3\text{CH}(\text{NH}_2)\text{CH}_3$ B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
 C) $\text{CH}_3\text{CH}_2\text{NHCH}_3$ D) $\text{CH}_3\text{CH}_2\text{CH}_3$

SECTION-III

(NON-NEGATIVE INTEGER.)

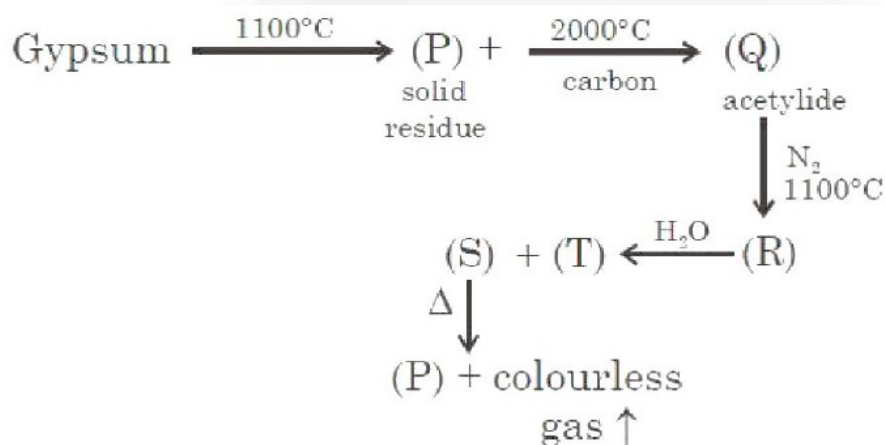
This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

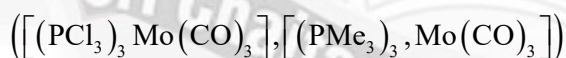
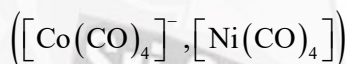
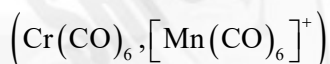
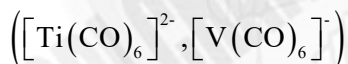
42. KMnO_4 oxidizes X^{n+} ion to XO_3^- in acid solution. 2.5×10^{-3} mole of X^{n+} requires 1.5×10^{-3} mole of MnO_4^- . How many moles of KMnO_4 is needed per mole of X^{n+} to oxidize it to XO_3^- in strong basic medium?

43.



Find the sum of sigma and pi bonds in anionic part of (R).

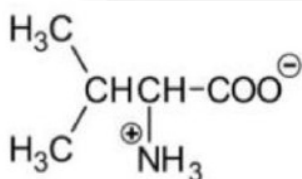
44. Find out the number of sets, in which first species would be expected to have the higher C-O stretching frequency?



45. The rate law for the reaction of OH^- with tert-butyl bromide to form an elimination product in 75% ethanol 25% water at 30°C is the sum of the rate laws for the E_2 and E_1 reactions: $\text{Rate} = 7.2 \times 10^{-5} [\text{tert-butyl bromide}] [\text{HO}^-] + 4.0 \times 10^{-5} [\text{tert-butyl bromide}]$

What percentage of reaction takes place by E_2 path way when conc. of $[\text{HO}^-]$ is 5.0M.

46. In order to estimate O_3 in the air, 22.4L of air at 1 atm and 0°C , is passed through acidified KI solution where O_2 is evolved and I^- is oxidized to I_2 . The liberated I_2 required 200 mL of 0.1 M $Na_2S_2O_3$ in acidic medium, then what will be the % composition of O_3 by volume in the sample?
47. The pH at the half-equivalence point of titration of the Zwitter ion of valine is 9.714 with NaOH and 2.286 with HCl, Thus, its isoelectric pH is



SECTION – IV (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

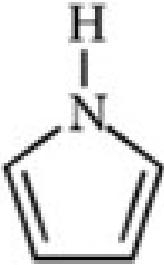
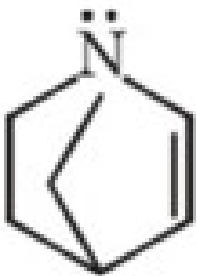
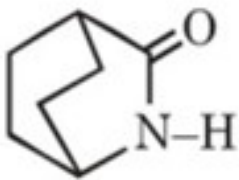
Negative Marks: -1 In all other cases..

48. Match the column

	List - I		List - II
A	$[Pt(NH_3)_2Cl_4]$	P	Complex molecule/ion in which neutral ligand is present
B	$[Cr(en)_2Cl_2]^+$	Q	Complex which shows geometrical isomerism
C	$[Cr(gly)_3]$	R	d^2sp^3 – hybridization is present in central atom of complex molecule/ion
D	$[Cr(en)_3]^{3+}$	S	Two geometrical isomers
		T	Complex in which only nitrogen atom acts as donor atom

- A) $A \rightarrow p, q, s; B \rightarrow p, q, r; C \rightarrow q, r, s; D \rightarrow p, q, t$
- B) $A \rightarrow p, q, r, s; B \rightarrow p, q, r, s; C \rightarrow q, r, s; D \rightarrow p, r, t$
- C) $A \rightarrow p, q, r; B \rightarrow p, q, r; C \rightarrow q, r; D \rightarrow p, q, t$
- D) $A \rightarrow p, q, r; B \rightarrow p, q, r; C \rightarrow p, q, r; D \rightarrow q, s$

49. Resonance, hyper conjugation and inductive effects, affect chemical properties of molecule/ion. These electronic effects, affect electron density at different centres in molecules. Now consider molecules given in List-I with their properties in List –II

	List-I (Species)		List-II (Properties)
(I)		(P)	Can undergo resonance
(II)	$H_3C - CH = CH - \overset{\overset{O}{\parallel}}{C} - CH_3$	(Q)	Can undergo hyperconjugation
(III)		(R)	Can act as electrophile
(IV)		(S)	Can act as Nucleophile
		(T)	Can undergo tautomerization
		(U)	Can react with Grignard reagent

Which of the following option contains all correct combinations?

- A) I – P, II – R, III – T, IV – Q B) I – U, II – Q, III – S, IV – Q
C) I – S, II – T, III – P, IV – T D) I – P, II – R, III – S, IV – U

50. Match the columns

	List - I		List - II
A		P	Formation of six member ring takes place
B		Q	Final product is ketone
C		R	Final product formed will give positive Tollen's test
D		S	Final product formed will react with 2, 4-DNP (2, 4- di-nitrophenyl hydrazine)

A) $A \rightarrow p, q, s; B \rightarrow p, q, s; C \rightarrow p, q, s; D \rightarrow p$

B) $A \rightarrow p, q, s; B \rightarrow p, q, s; C \rightarrow p; D \rightarrow p, q, r$

C) $A \rightarrow p; B \rightarrow p, q, r; C \rightarrow p, q, s; D \rightarrow p, q, s$

D) $A \rightarrow p, q, s; B \rightarrow p; C \rightarrow p, q, s; D \rightarrow p, q, s$
51. Match the items of Columns I and II and mark the correct option

	List - I		List - II
A	Solid dispersed in liquid	P	Emulsion
B	Liquid dispersed in liquid	Q	Foam
C	Gas dispersed in liquid	R	Gel
D	Liquid dispersed in Solid	S	Solution

A) $a-s, b-p, c-q, d-r$

B) $a-p, b-r, c-q, d-s$

C) $a-s, b-q, c-r, d-p$

D) $a-p, b-q, c-r, d-s$



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RANK 37 SUMEDH SS HT.No. 231028088	RANK 38 V DEERAJ SATHVIK REDDY HT.No. 230050751	RANK 44 V VERGIYA KAUSHAL HT.No. 231004059	RANK 46 SUTHAR HARSHAL HT.No. 231005060	RANK 58 ABHI JAIN HT.No. 232091036	RANK 61 RIDAM JAIN HT.No. 232068021	RANK 62 SUTHAR HARSH HT.No. 231000028	RANK 63 SOJITRA DIPEN HT.No. 231077001	RANK 65 D PANENDRANADHA REDDY HT.No. 236050298
RANK 67 GUNDA SUSHANTH HT.No. 236128732	RANK 68 HARSH TAYA HT.No. 237021178	RANK 69 KRISH GUPTA HT.No. 232088189	RANK 73 KUNCHAKURTHY VIVEK HT.No. 230134083	RANK 81 N DHARMA TEJA REDDY HT.No. 236050138	RANK 84 S VENKAT KOUNDINYA HT.No. 236130679	RANK 87 MRUNAL SHRIKANT HT.No. 23104138	RANK 92 SAMOTA APURVA HT.No. 232098020	RANK 97 N PRINCE BRANHAM REDDY HT.No. 236064112

BELOW 20
All India Open
Category Ranks

10
RANKS (50%)

BELOW 100
All India Open
Category Ranks

32

BELOW 1000
All India Open
Category Ranks

181

BELOW 100
All India
Category Ranks
Count

89

BELOW 1000
All India
Category Ranks
Count

699

NUMBER OF RANKS QUALIFIED
3,621

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